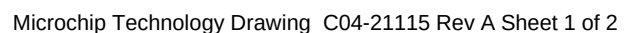
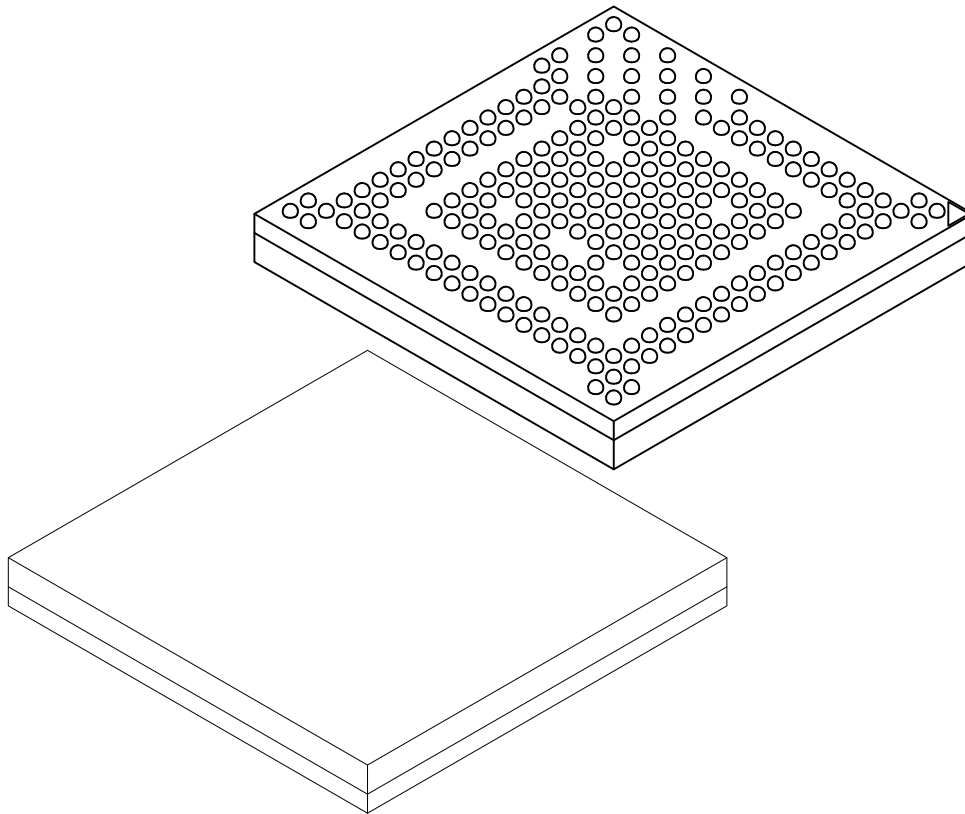


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



247-Ball Low Profile Fine Pitch Ball Grid Array (A7B) - 10x10x1.37 mm Body [LFBGA] Atmel Legacy Global Package Code CFR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



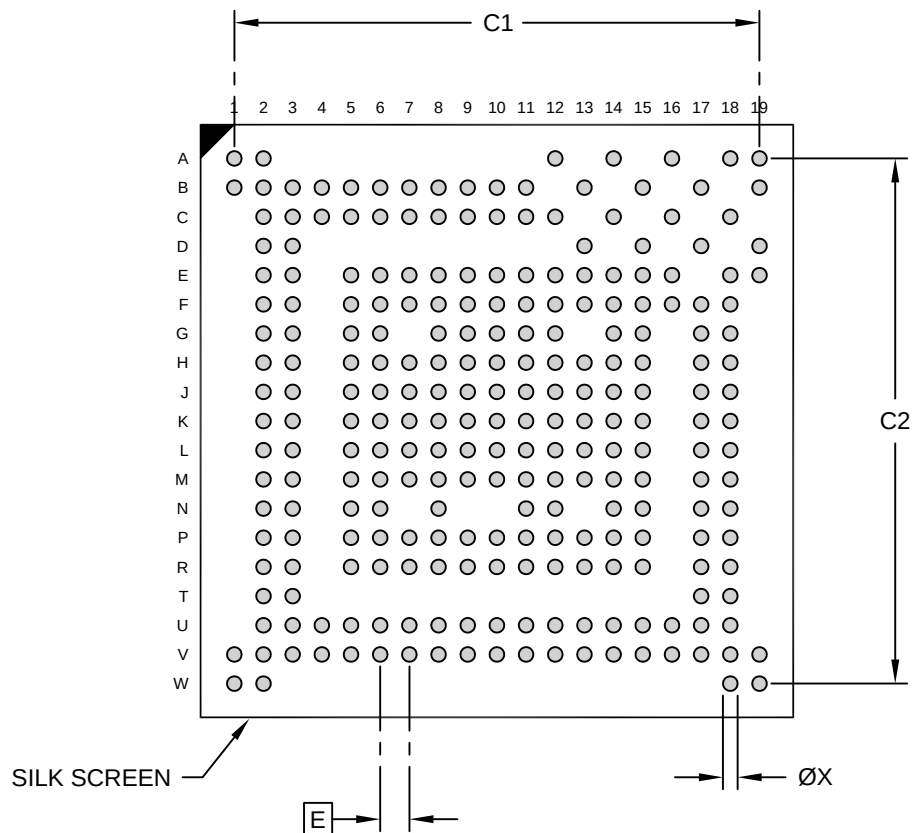
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	247		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.37
Ball Height	A1	0.16	0.21	0.26
Molded Package Height	A2	1.02	1.06	1.10
Mold Cap Height	A3	0.70 REF		
Overall Length	D	10.00 BSC		
Overall Width	E	10.00 BSC		
Terminal Diameter	b	0.25	0.30	0.35

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

247-Ball Low Profile Fine Pitch Ball Grid Array (A7B) - 10x10x1.37 mm Body [LFBGA] Atmel Legacy Global Package Code CFR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		9.00	
Contact Pad Spacing	C2		9.00	
Contact Pad Diameter (X247)	X2			0.25

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23115 Rev A